# CTA200H Assignment 3

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## 1 Introduction

## 2 Methods

### 2.1 Complex Plane Iteration; Mandelbrot Set

The Mandelbrot Set is an famous set of numbers in the complex plane that produce a "fractal" (a shape with infinite perimeter and finite area). Essentially, in a complex plane where the boundaries are

$$-2 < x < 2 \text{ and} - 2 < y < 2 \tag{1}$$

there are a set of points that do not diverge after a set number of iterations of the following formulae:

$$c = x + iy \tag{2}$$

$$z_{i+1} = z_i + c \tag{3}$$

Where $z_n$ is the $n^{th}$ iteration of equation 2.

Using numpy, we can perform the iteration for points within the boundary and observe whether or not they diverge within a fixed number of iterations. We then plotted the points all points within the boundary by using the matplotlib.pyplot function, "imshow", which displays points in a complex plane. From these points, two plots were generated; one that showed which points diverged, and another that showed how many iterations it took for divergence to occur (or if the maximum number of iterations was reached).

### 2.2 Deterministic Modeling of Earth's Atmosphere

From Lorenz's paper, we recreated his modelling of the Earth's atmosphere considering thermal convection and fluid viscosity. The equations to model this are:

$$\dot{X} = -\sigma(X - Y) \tag{4}$$

$$\dot{Y} = rX - Y - XZ \tag{5}$$

$$\dot{Z} = -bZ + XY \tag{6}$$

Using the solve_ivp function from the scipy.integrate library, we can find a solution to this ODE, and observe the behaviour of each parameter (X, Y, and Z) over a number of iterations. Furthermore, we can use numpy to get slighly altered initial conditions, where W' = (0., 1.00000001, 0.), instead of W = (0, 1, 0). Then, we find the distance between W' and W over time.

# 3 Results

## 3.1 Mandelbrot Plots

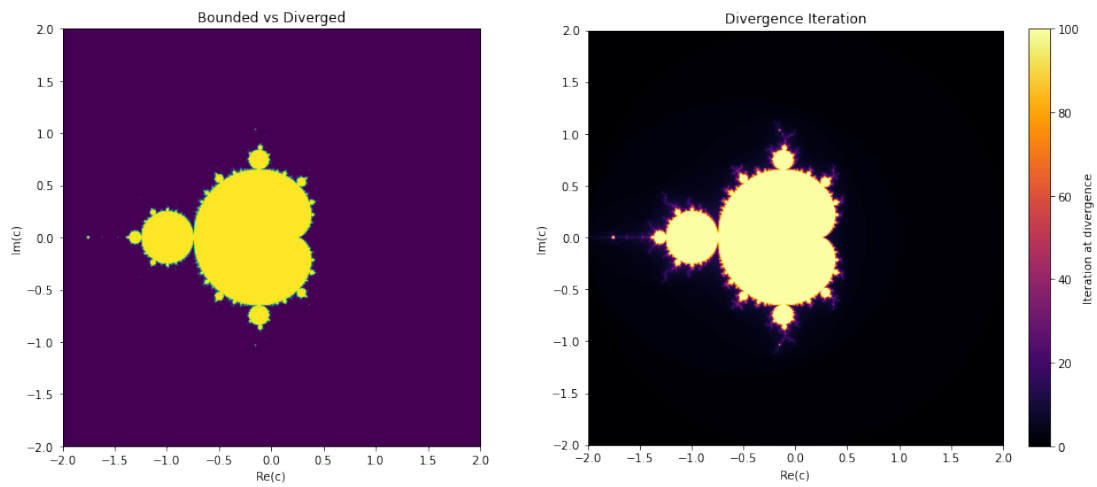


Figure 1: Mandelbrot Plots with boundaries; $-2 < x < 2$ and $-2 < y < 2$

Here, we can observe that there is a coherent "fractal" shape that is formed along the boundary of points that diverge (purple), and do not diverge (yellow) after a number of iterations. In the second plot, we can observe a few distinct features. The black represents points that diverge above —2— within a small number of iterations (¡¡10), whereas the "warmer" colours represent points that diverge in a greater number of iterations. Finally, the yellow points within the fractal are points that do not diverge after 100 iterations.

## 3.2 Lorenz Plots

In this figure, we can observe various epochs within the plot. Initially, there is a dramatic spike, before it oscillates. After roughly 1600 iterations, there seem to be numerous, ordered, but significant spikes

We can also observe cyclic behaviour over time in the X-Y and Y-Z plots above. In the above plot, we measured the distance between W' and W.

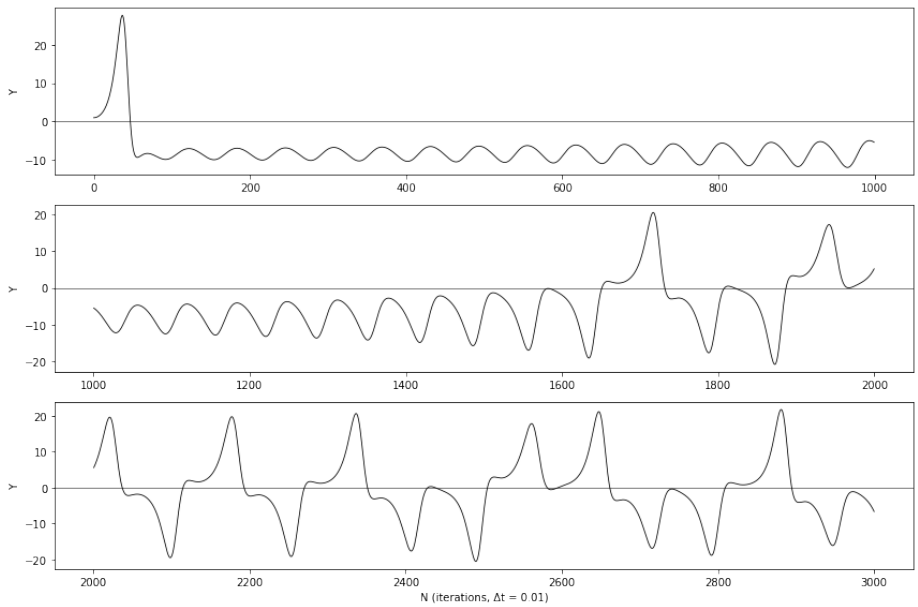


Figure 2: Recreated Figure 1 of Lorenz's plots. The x-axis represents the number of iterations, and the y-axis is the corresponding y-value

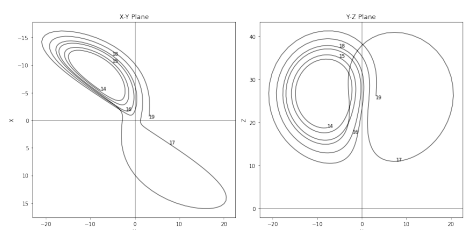


Figure 3: X-Y and Y-Z plots

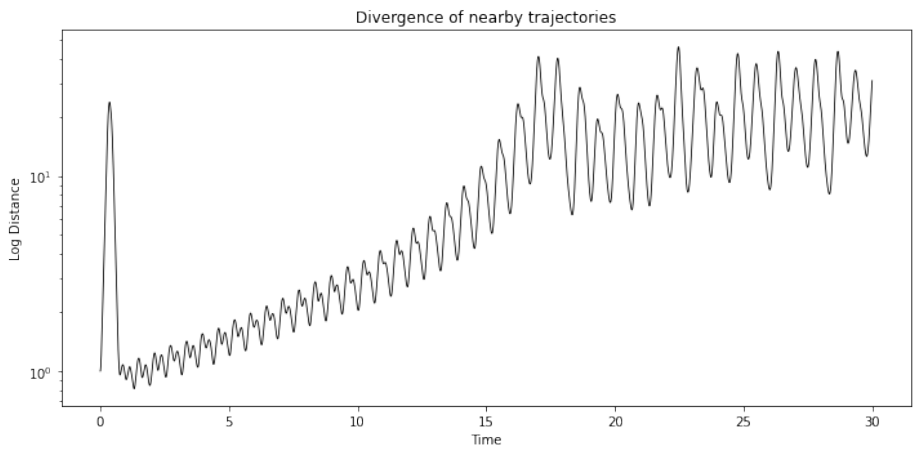


Figure 4: W' distance plots